

END-TO-END DATA PROCESSING & VISUALIZATION

using Python

Case Study About E-Commerce User Behavior Analysis



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01

BUSINESS CONTEXT & OBJECTIVES



Role

Associate **Data Analyst**



Company: BukaToko

E-commerce platform operating across Southeast Asia, US, and Australia.



Goal

- Explore **user behavior**
- Provide **data-driven insights** to optimize Marketing Strategy

OBJECTIVES

1

Market Identification

Identify the country with the **highest user activity** in Q2 2025

2

Growth Trend Analysis

Analyze **Monthly Active User** trends across different acquisition channels

3

Device Preference

Determine which device is **most frequently used** by customers

4

Feature Effectiveness

Evaluate the "**Search-to-Cart**" **conversion rate** to measure search feature performance

5

Peak Activity Analysis

Identify the **busiest days and hours** to optimize engagement timing

DATA PREPARATION & QUALITY CHECK

```
import pandas as pd
import numpy as np

spreadsheet_url = "https://docs.google.com/spreadsheets/d/1dDczkp5SRp6IMNpKxiak_4q0S5qr0SILiYKzp-2I9ZE/export?format=csv&gid=1307725687"
df = pd.read_csv(spreadsheet_url)

df.head()
```

	event_id	user_id	session_id	event_type	event_timestamp	product_id	country	device	channel
0	9bd8725f-5722-41b7-8d95-a5dafec33a56	482c0ae4-ac6c-4eb3-91fe-b0e63c1c2a5d	98b232db-5bec-41d8-88e1-5d2e98899e29	product_view	2025-09-22 16:35:27	P1208	US	IOS	paid_search
1	f6dc704c-a88c-4668-b89d-e6b08e7e02d4	ff6c808d-fa75-4202-801f-a076299a2c0f	d4832af2-0aad-489c-8eb6-22d243f5f5c4	page_view	2025-06-04 21:55:04	NaN	ID	android	referral
2	abb69be8-df83-420b-92f7-149d29a6d32f	78ea2c10-e824-4b3c-b74b-cdb0cbab0610	1f6c5b12-6c79-4b29-bae6-878804723e55	page_view	2025-06-24 2:02:51	NaN	AU	IOS	paid_search
3	76fd2399-0f80-4184-8802-90dac9bf52d3	9c63419c-b3dc-44a6-b8a5-a438e82afee8	ed1c4ded-cf67-463e-b94f-4cebcf62d827	page_view	2025-06-05 12:35:33	NaN	VN	IOS	organic
4	26dcc5fa-334c-420c-be1e-a40eb4b9efac	f9aee3bb-2ba4-4bb3-9eec-44ba4664f93b	9a469fb9-6cf4-4818-b0a6-7302b0bd665a	product_view	2025-07-02 12:19:23	P1315	ID	desktop	social

[Click here to view dataset](#)

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   event_id              10000 non-null  object
1   user_id              10000 non-null  object
2   session_id           10000 non-null  object
3   event_type            10000 non-null  object
4   event_timestamp       10000 non-null  object
5   product_id           4463 non-null   object
6   country              10000 non-null  object
7   device               10000 non-null  object
8   channel              9900 non-null   object
dtypes: object(9)
```

event_timestamp was stored as an object (string)

Detected Null values in the channel column.

ACTION

Converted column to datetime format

```
df["event_timestamp"] = pd.to_datetime(df["event_timestamp"])
```

```
df["date"] = df["event_timestamp"].dt.date
```

Extracted the date component to a new column

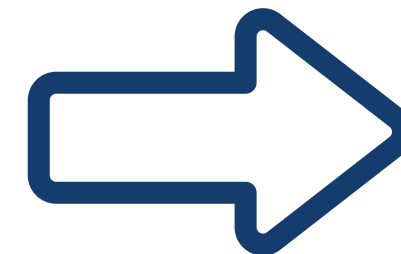
```
df["channel"] = df["channel"].fillna("organic")
```

Imputed missing values with "organic" (assuming direct traffic)

AFTER

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   event_id              10000 non-null  object
1   user_id              10000 non-null  object
2   session_id           10000 non-null  object
3   event_type            10000 non-null  object
4   event_timestamp       10000 non-null  datetime64[ns] ✓
5   product_id           4463 non-null   object
6   country              10000 non-null  object
7   device               10000 non-null  object
8   channel              ✓10000 non-null  object
9   date                 ✓10000 non-null  object
dtypes: datetime64[ns](1), object(9)
```





Feature Effectiveness

Evaluate the "Search-to-Cart" conversion rate to measure search feature performance

event_type
product_view
page_view
page_view
page_view
product_view
purchase
product_view
login
page_view
checkout
search
page_view
add_to_cart
add_to_cart



In the raw dataset, **search** and **add_to_cart** are recorded as separate rows.



We must strictly identify users who performed *both* actions **on the same calendar date**.

```
search_data = df[df['event_type'] == 'search'][['user_id', 'date']].drop_duplicates()
```

```
atc_data = df[df['event_type'] == 'add_to_cart'][['user_id', 'date']].drop_duplicates()
```



```
converted_data = pd.merge(search_data, atc_data, on=['user_id', 'date'], how='inner')
```

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ANALYTICAL LOGIC: SEARCH-TO-CART ATTRIBUTION

Step 1: Isolate & De-duplicate

Separate 'Search' and 'ATC' events and remove **duplicates**.

Step 2: 'Same Day' Logic

Use **Inner Join** to find users matching both actions **on the same date**

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OBJECTIVE #1: MARKET IDENTIFICATION

THE CODE

```
df['month_name'] = df['event_timestamp'].dt.month_name()
q2_months = ['April', 'May', 'June']
df_q2 = df[df['month_name'].isin(q2_months)]
top_country = df_q2.groupby('country')['user_id'].nunique().sort_values(ascending=False)
```

Converted column to **datetime format**

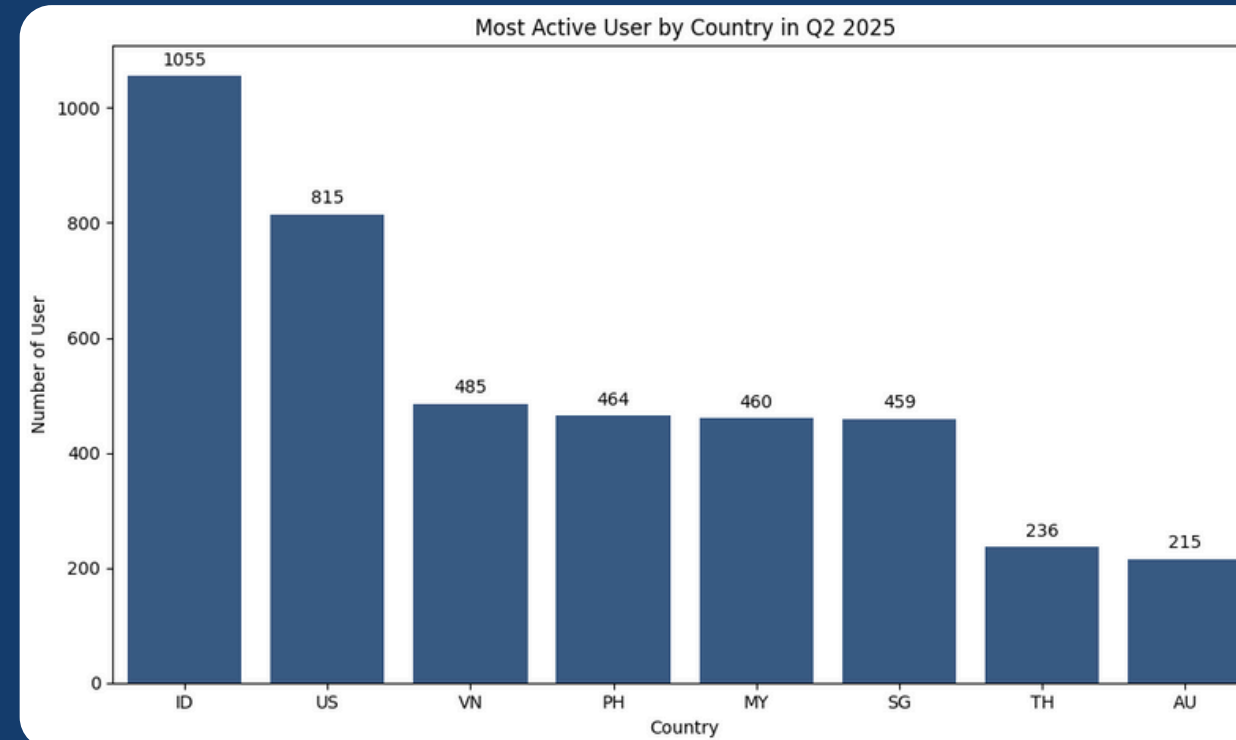
Isolating data strictly for the requested period (**April-June**)

Filters dataset to include only **q2_months**

Groups data by country

Counts distinct users (avoids double-counting the same person)

Ranks results **from highest to lowest** activity

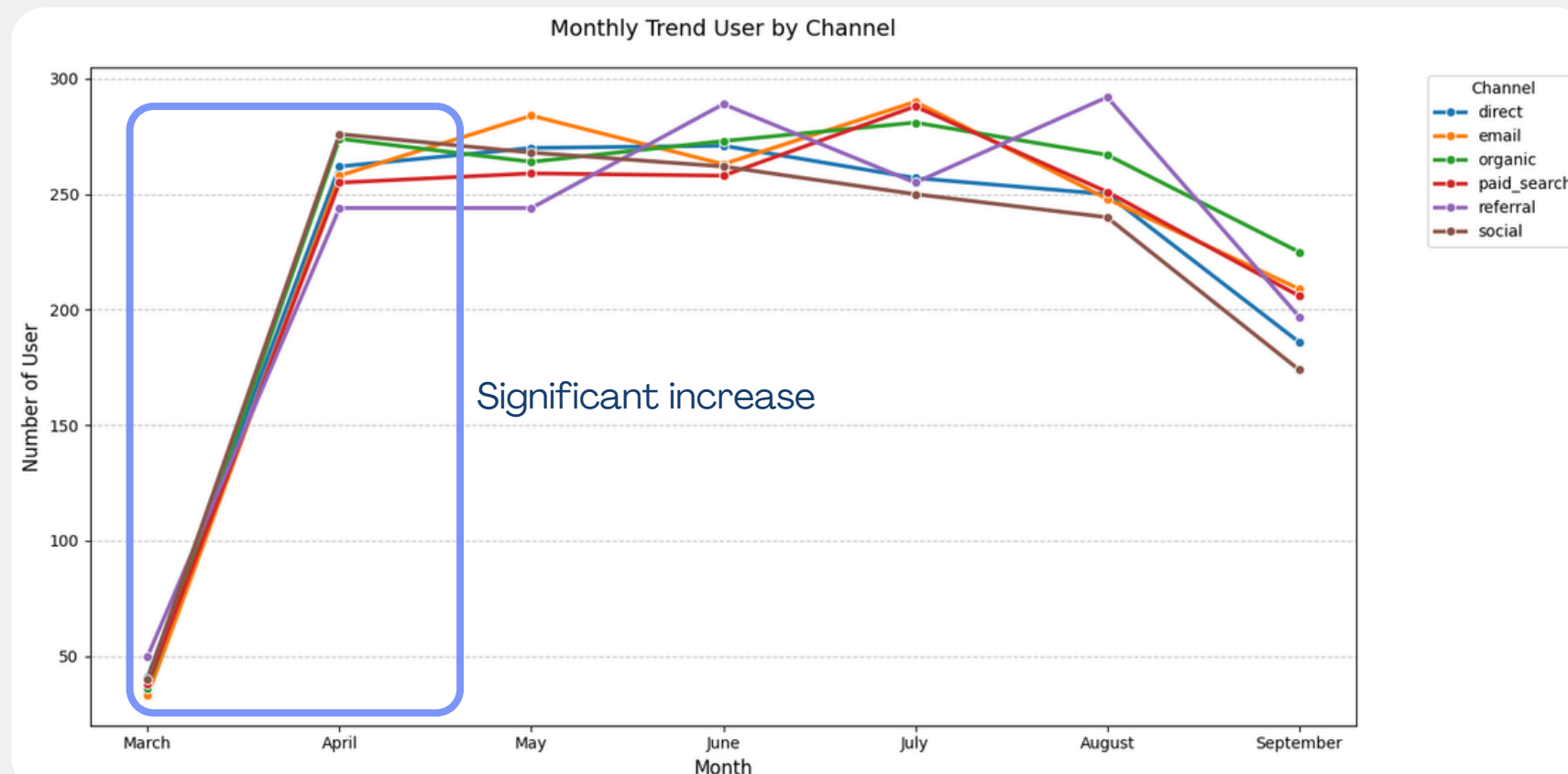


**Q2 MARKET
FOCUS:
INDONESIA
LEADS IN
USER
ACTIVITY**

Indonesia (ID) is the most active market with **1,055** users, significantly outperforming the **US (815)** and **Vietnam (485)**

CHANNEL PERFORMANCE OVERVIEW

Analysis reveals that recorded data is available from **March to September** 2025. A coordinated increase **occurred in April** across all channels, followed by stability until August, with a **significant decline beginning in September**.



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OBJECTIVE #2: GROWTH TREND

```
channel_trend = df.pivot_table(
    index='month_name',
    columns='channel',
    values='user_id',
    aggfunc='nunique'
)
```

Reshapes data for **multi-line comparison**

Ensures accuracy by **counting unique users**

```
months_order = ['January', 'February', 'March', 'April', 'May', 'June', 'July',
                'August', 'September', 'October', 'November', 'December']
```

```
channel_trend = channel_trend.reindex(months_order)
```

Fixes sorting to **chronological order**
(Prevents April → August)

```
channel_trend = channel_trend.dropna(how='all')
```

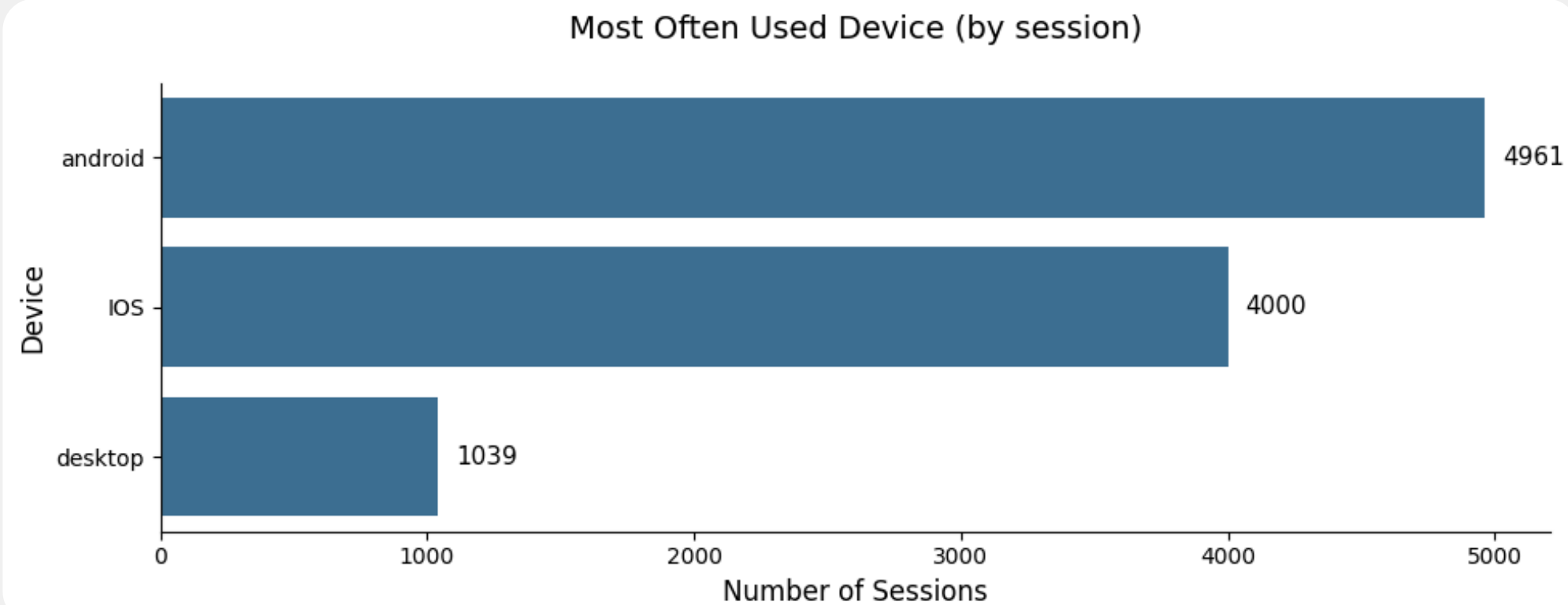
Delete rows where the **entire month is NaN**

```
display(channel_trend)
```

Table

channel	direct	email	organic	paid_search	referral	social
month_name						
March	41.0	33.0	36.0	38.0	50.0	40.0
April	262.0	258.0	274.0	255.0	244.0	276.0
May	270.0	284.0	264.0	259.0	244.0	268.0
June	271.0	263.0	273.0	258.0	289.0	262.0
July	257.0	290.0	281.0	288.0	255.0	250.0
August	250.0	248.0	267.0	251.0	292.0	240.0
September	186.0	209.0	225.0	206.0	197.0	174.0

OBJECTIVE #3: DEVICE PREFERENCE



```
users_per_device = df.groupby('device')['session_id'].nunique().sort_values(ascending=False)
```

Categorizing interactions to identify the dominant platform used by customers.

MOBILE-FIRST BEHAVIOR

The majority of access comes from **Android & iOS**, indicating a strong dependency on mobile apps **compared to Desktop/Web**

```
display(users_per_device)
```

session_id	
device	
android	4961
ios	4000
desktop	1039

07

OBJECTIVE #4 FEATURE EFFECTIVENESS

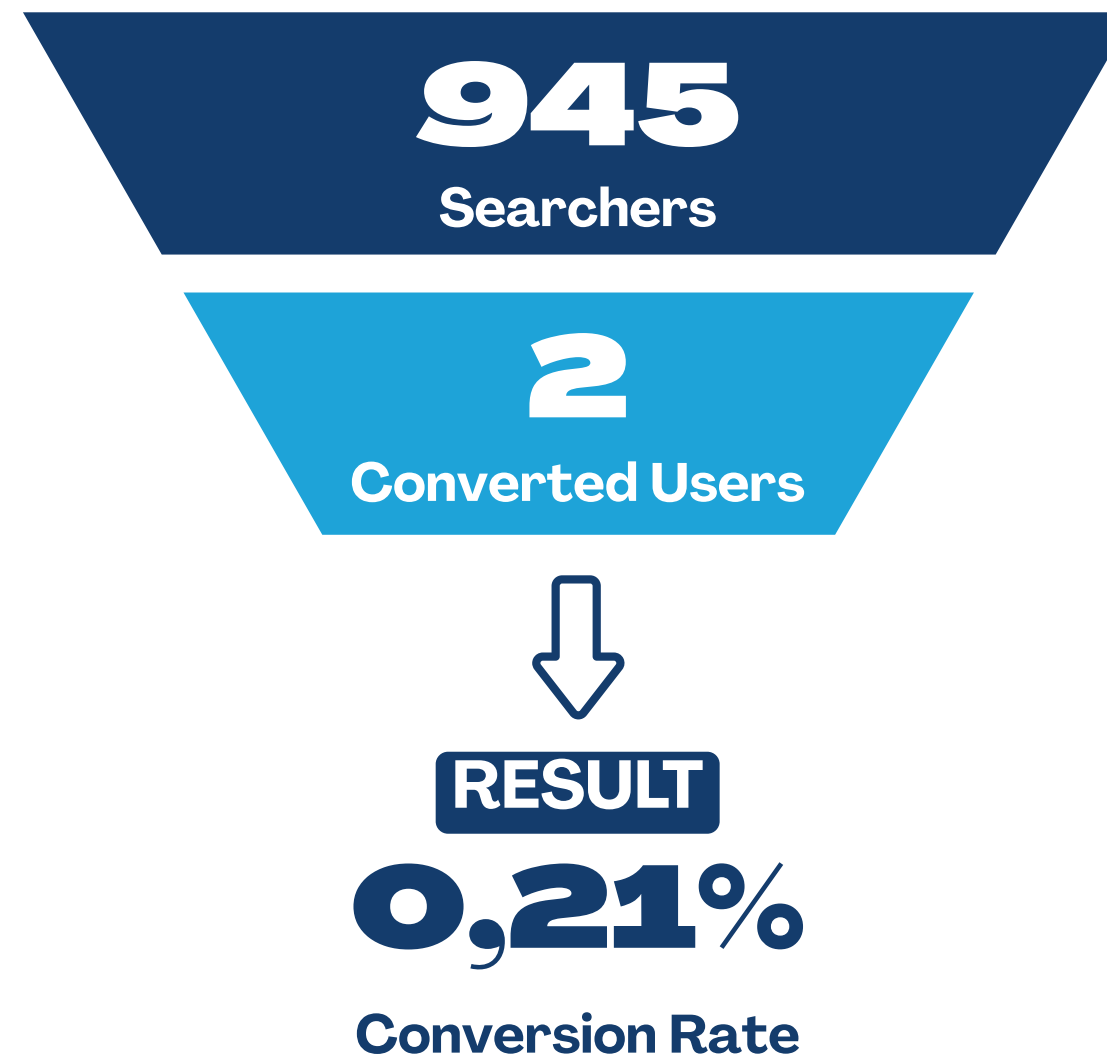
Applying the "Search-to-Cart Attribution Logic" established in the Methodology section (Point 03), now we quantify the actual conversion performance.

```
total_searchers = len(search_data)
total_converted = len(converted_data)
```

→ Extracting User Counts

```
conversion_rate = (total_converted / total_searchers) * 100
```

→ Conversion formula



Converted Data Table

	user_id	date
0	4074cba6-8de9-4660-b191-e08fc7bf6dc2	2025-05-13
1	3bd58b47-09d8-473d-ac36-19578b89edfb	2025-08-07

The 0.21% Conversion Challenge

The conversion rate is **critically low**. This confirms that while users utilize the Search feature, immediate **intent to purchase is almost non-existent**.

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OBJECTIVE #5: ACTIVITY ANALYSIS

```
df['hour'] = df['event_timestamp'].dt.hour  
df['day_of_week'] = df['event_timestamp'].dt.day_name()
```

Extracts **time components** (Hour & Day)

```
day_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']  
df['day_of_week'] = pd.Categorical(df['day_of_week'], categories=day_order, ordered=True)
```

Enforces **chronological sorting** (Mon -> Sun) to **prevent default alphabetical ordering**

```
activity = df.groupby(['day_of_week',  
                        'hour'])['user_id'].nunique().unstack(fill_value=0)
```

Transforms data into a matrix format (**Rows=Days, Cols=Hours**)

```
display(activity)
```

hour	0	1	2	3	4	5	6	7	8	9	...	14	15	16	17	18	19	20	21	22	23				
day_of_week	Monday	60	57	61	75	57	73	66	78	54	52	61	70	58	60	54	60	64	58	51	48	47	69	50	59
Tuesday	58	65	66	63	47	57	64	56	61	61	53	58	65	58	54	61	56	64	58	48	39	61	57	64	
Wednesday	51	48	56	64	62	48	60	63	63	50	68	63	45	60	53	60	49	52	70	68	61	57	52	66	
Thursday	50	59	61	45	60	51	62	53	52	54	60	43	63	73	56	59	64	62	62	58	54	55	50	64	
Friday	47	70	66	56	61	56	64	49	53	56	51	66	56	61	59	50	55	56	65	55	53	52	58	67	
Saturday	60	61	65	72	58	60	56	77	57	58	50	63	56	45	51	54	53	70	48	52	75	56	58	56	
Sunday	77	57	67	57	56	61	57	64	65	68	65	59	54	74	55	42	62	58	52	62	67	55	65	64	

7 rows x 24 columns

OPTIMIZATION WINDOW

User activity is well-distributed, but **Peak Engagement Windows** are identified during **commuter hours (7-8 AM & 5-7 PM)**.

STRATEGIC ROADMAP: 3 KEY PILLARS

1

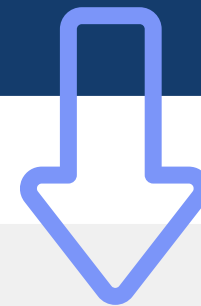
Indonesia dominates traffic, and users are heavily mobile-centric



Allocate Q3 engineering resources to optimize Mobile App performance specifically for the Indonesian region. Adopt a "Mobile-First" design for all new features.

2

Search-to-Cart conversion is critically low (0.21%)



Audit the search algorithm relevancy and pricing competitiveness. Conduct A/B testing on the search result page (UI/UX) to reduce drop-off.

3

Commuter hours (7-8 AM & 5-7 PM) are peak times, but a seasonal slump is predicted for September



Schedule push notifications during commuter windows. Launch a "September Super Sale" or loyalty campaign in late August to prevent the projected traffic decline

THANK YOU

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